

MONITORING AND CONTROLLING SERVICE PERFORMANCE AND EFFICIENCY IN DYNAMIC DATACENTERS

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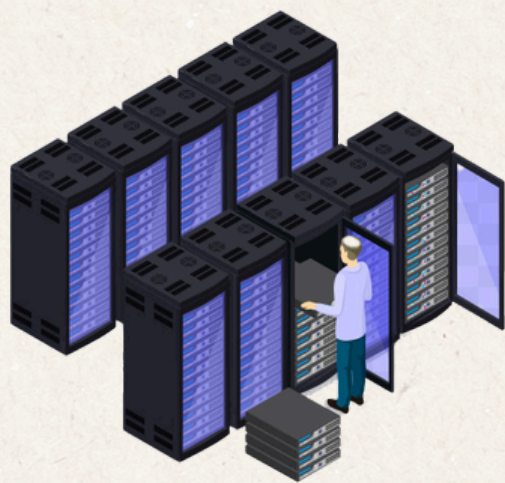
Context

The growing complexity of distributed datacenter infrastructures demands intelligent monitoring frameworks capable of providing multi-layer observability, fault/stress testing, and data-driven performance analysis.



01

Functional Requirements



Collect infrastructure metrics

Centralize and store monitoring data

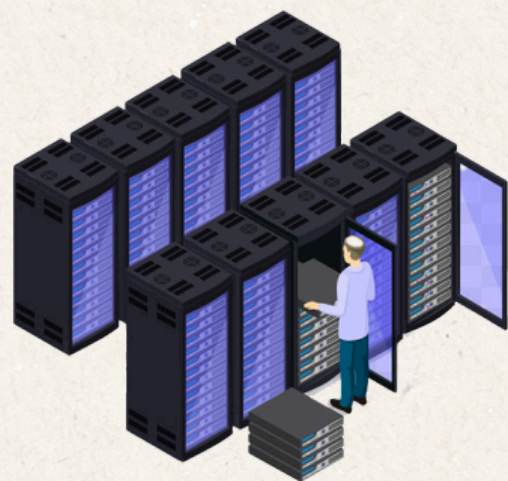
Perform fault injection and stress testing

Analyze data using ML/anomaly detection

Provide dashboards and alerts for operators

02

Non-Functional Requirements



Data processing scalability
Monitoring infrastructure scalability
Low intrusiveness (Overhead)
Availability of monitorization pipeline
Latency of anomaly detection
ML Accuracy

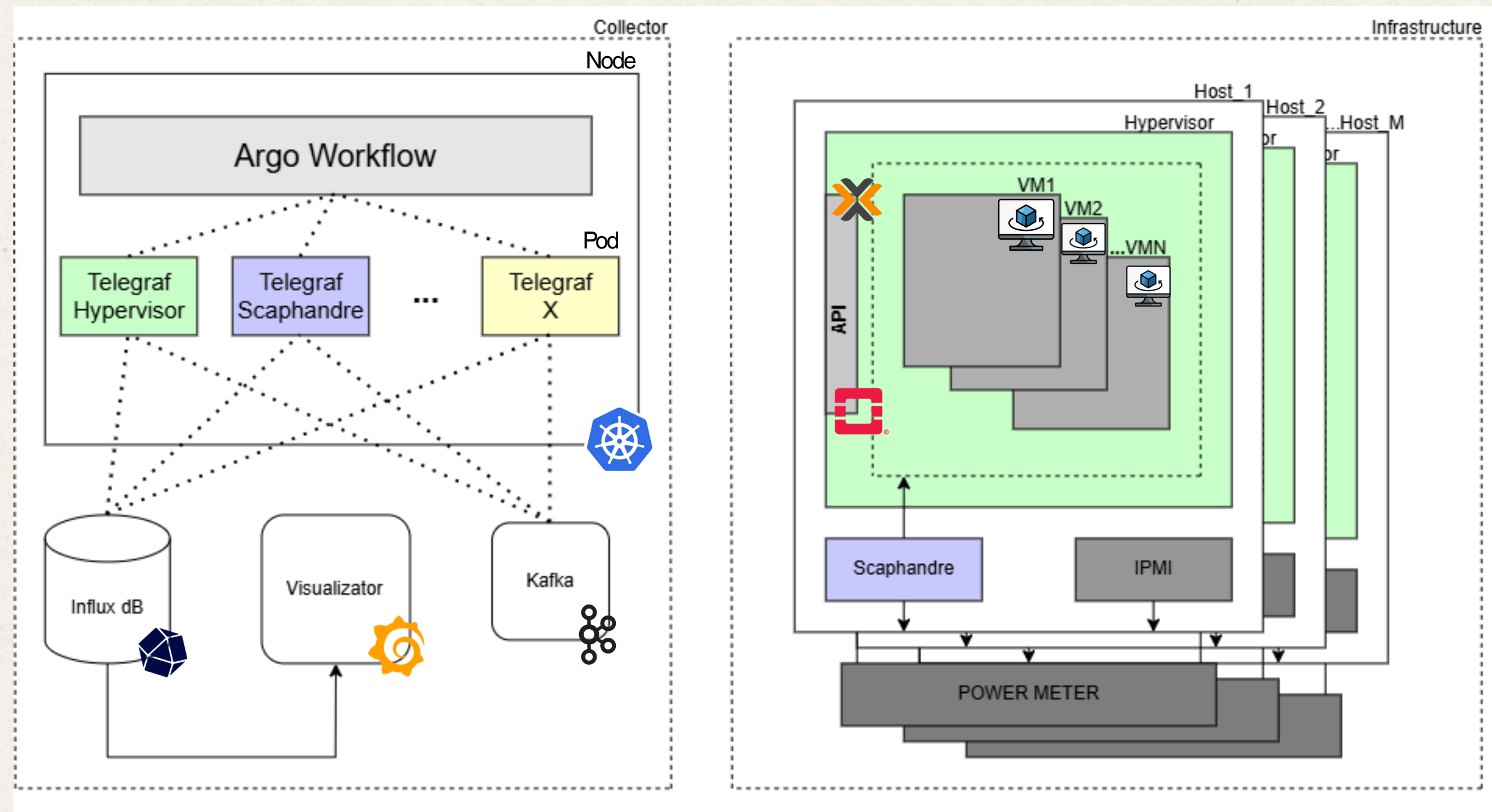
03

Architecture

- **Collector**

VM - Virtual
Machine

IPMI - Intelligent
Platform
Management
Interface

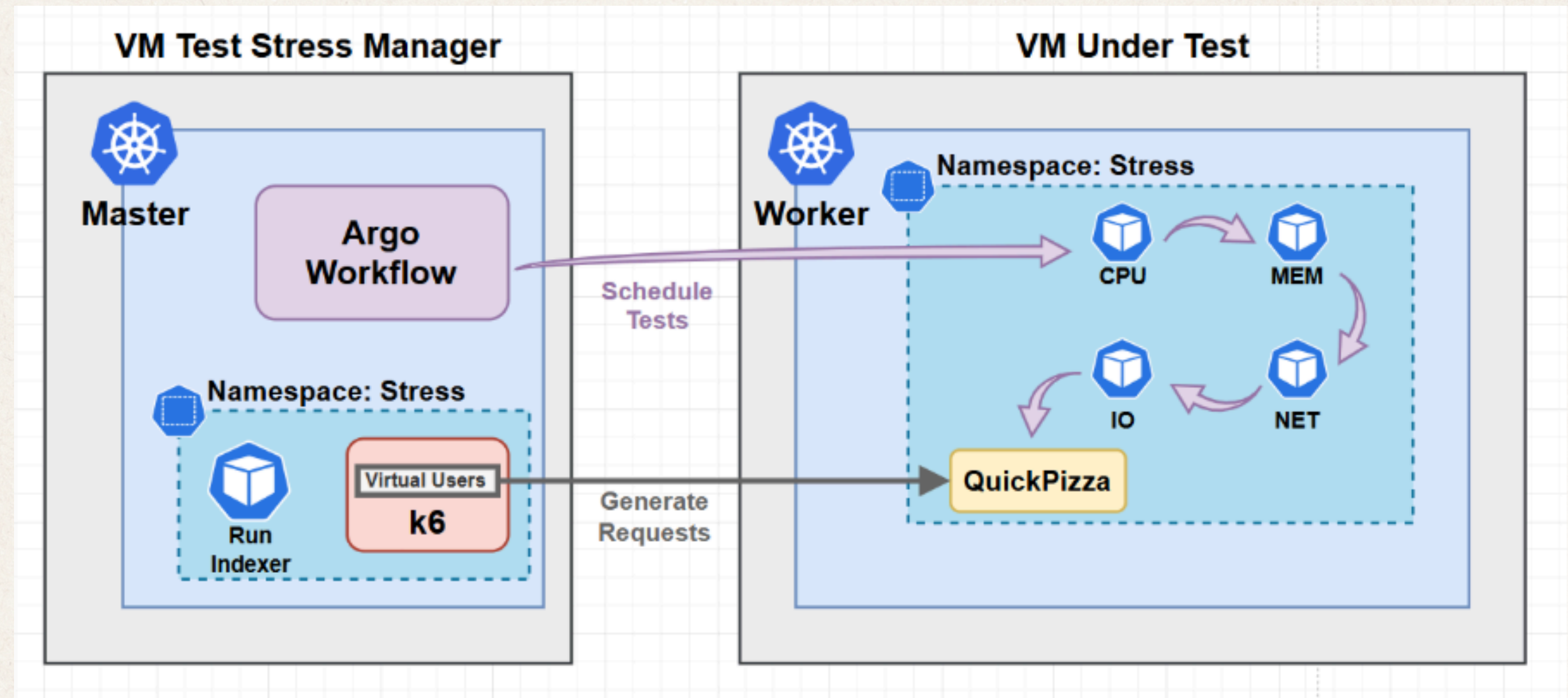


03

Architecture

- Stress Tests

VM - Virtual
Machine



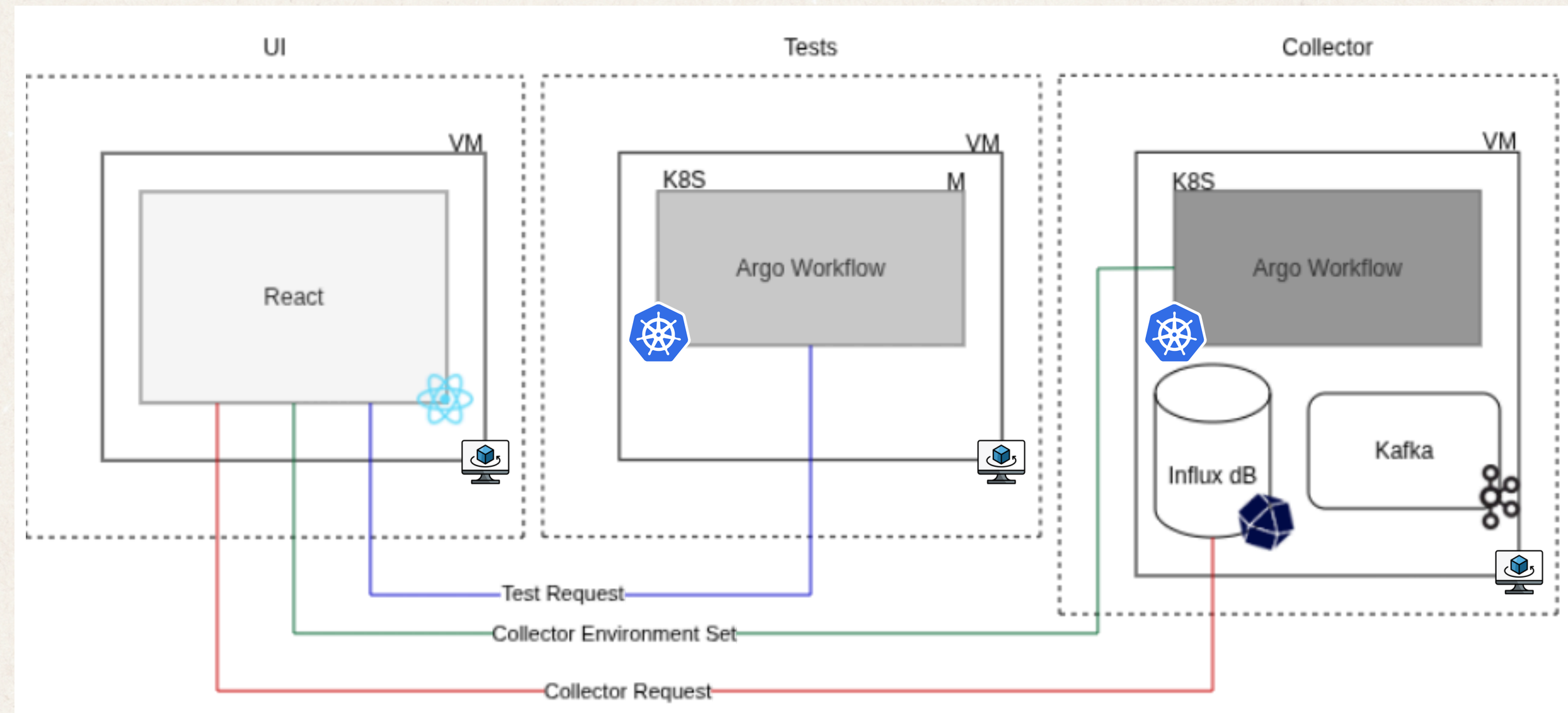
03

Architecture

• UI

VM - Virtual
Machine

M - Master



04

Demo

Dashboard

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